

TAXES AND SUBSIDIES PROBLEM SET—SUGGESTED SOLUTIONS

ECON101—INTRODUCTION TO MICROECONOMICS.

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Problem 1

- (a) $P^e = \frac{a-c}{b+d}$
- (b) $Q^e = \frac{ad+bc}{b+d}$
- (c) $P^d = \frac{a-c+dT}{b+d}$
- (d) $P^s = \frac{a-c-bT}{b+d}$
- (e) $Q^T = \frac{ad+bc-bdT}{b+d}$
- (f) Government revenue is $\frac{T(ad+bc-bT)}{b+d}$
- (g) Consumer incidence is $\frac{b}{a+b}$ and producer incidence is $\frac{a}{a+b}$
- (h) $DWL = \frac{bdT^2}{2(b+d)}$

Problem 2

- (a) $P^d = \frac{a-c+bS}{b+d}$
- (b) $P^s = \frac{a-c-dS}{b+d}$
- (c) $Q^S = \frac{ad+bc+bdS}{b+d}$
- (d) Government expenditure is $\frac{S(ad+bc+bdS)}{b+d}$
- (e) Consumer incidence is $\frac{d}{b+d}$ and producer incidence is $\frac{b}{b+d}$, similar to the case of a tax.
- (f) $DWL = \frac{bdS^2}{2(b+d)}$

Problem 3

- (a) If $b = 0$ then $P^s = P^e$, which implies that the consumer pays the whole tax/receives the whole subsidy, as can be seen by the consumer incidence expression. Furthermore Q^T and Q^S are equal to Q^e .
- (b) If $d = 0$ then $P^d = P^e$, which implies that the producer pays the whole amount of the tax/receives the whole subsidy, as can be seen by the producer incidence expression. Furthermore, the dead-weight loss in this case is also zero. Furthermore Q^T and Q^S are equal to Q^e .

- (c) In both the tax and subsidy cases, the dead-weight loss expression becomes zero if either $b = 0$ or $d = 0$. When demand or supply are perfectly inelastic, the distortion created by the tax or subsidy is zero and therefore the dead-weight loss is zero.
- (d) As mentioned before, when demand is inelastic, the tax incidence/subsidy incidence expression says that the consumer will pay the whole tax/receive the whole subsidy. Similarly, when supply is inelastic, the tax incidence/subsidy incidence expression says that the producer will pay the whole amount of the tax/receive the whole subsidy.

Problem 4

- (a) Consider first the case of perfectly elastic demand. We have that $Q^e = c + dP^*$ and $P^e = P^*$.

	Tax	Subsidy
P^d	P^*	P^*
P^s	$P^* - T$	$P^* + S$
$Q^T Q^S$	$c + dP^* - dT = Q^e - dT$	$c + dP^* + dS = Q^e + dS$
Revenue Expenditure	$T(c + dP^* - dT)$	$S(c + dP^* + dS)$
Consumer Incidence Producer Incidence	0 1	0 1
Dead-Weight Loss	$\frac{dT^2}{2}$	$\frac{dS^2}{2}$

- (b) Now consider the case of perfectly elastic supply. We have that $Q^e = a - bP^*$ and $P^e = P^*$.

	Tax	Subsidy
P^d	$P^* + T$	$P^* - S$
P^s	P^*	P^*
$Q^T Q^S$	$a - bP^* - bT = Q^e - bT$	$a - bP^* + bS = Q^e + bS$
Revenue Expenditure	$T(a - bP^* - bT)$	$S(a - bP^* + bS)$
Consumer Incidence Producer Incidence	1 0	1 0
Dead-Weight Loss	$\frac{bT^2}{2}$	$\frac{bS^2}{2}$

Problem 5

Table 1 Provides the calculations for taxes while table 2 shows the corresponding calculations for subsidies.

From both tables you can check that the identities indeed hold. Consider the following:

- For a tax, where $SS^{\text{before tax}} \geq SS^{\text{after tax}}$, the change in social surplus is $\Delta SS = SS^{\text{before tax}} - SS^{\text{after tax}}$. Thus, using the identity, we can write

$$\Delta SS - DWL = R,$$

which shows that government revenues are given by the change in social surplus minus the dead-weight loss which shows that indeed some part of the surplus is lost irrevocably.

Table 1: Calculations for taxes

	Case I	Case II	Case III	Case IV	Case V
Amount of tax paid by consumers	111.11	1.00	0.00	22,500.00	7.00
Amount of tax paid by producers	222.22	3.00	4,500,000.00	22,500.00	0.00
Government revenue from the tax	333.33	4.00	4,500,000.00	45,000.00	7.00
Consumer surplus before tax	400.00	2.00	81,000,000.00	45,000.00	32.00
Producer surplus before tax	750.00	6.00	9,000,000.00	45,000.00	0.00
Social surplus before tax	1,150.00	8.00	90,000,000.00	90,000.00	32.00
Consumer surplus after tax	277.78	0.50	81,000,000.00	11,250.00	24.50
Producer surplus after tax	505.56	1.50	4,500,000.00	11,250.00	0.00
Social surplus after tax	783.33	2.00	85,500,000.00	22,500.00	24.50
Dead Weight Loss	33.33	2.00	0.00	22,500.00	0.50

Table 2: Calculations for subsidies

	Case I	Case II	Case III	Case IV	Case V
Amount of subsidy received by consumers	155.56	2.06	0.00	100,000.00	180.00
Amount of subsidy received by producers	311.11	6.19	1,800,000.00	100,000.00	0.00
Government expenditure from the subsidy	466.67	8.25	1,800,000.00	200,000.00	180.00
Consumer surplus before subsidy	400.00	2.00	81,000,000.00	45,000.00	32.00
Producer surplus before subsidy	750.00	6.00	9,000,000.00	45,000.00	0.00
Social surplus before subsidy	1,150.00	8.00	90,000,000.00	90,000.00	32.00
Consumer surplus after subsidy	544.44	3.78	81,000,000.00	125,000.00	162.00
Producer surplus after subsidy	1,038.89	11.34	10,800,000.00	125,000.00	0.00
Social surplus after subsidy	1,583.33	15.13	91,800,000.00	250,000.00	162.00
Dead Weight Loss	33.33	1.13	0.00	40,000.00	50.00

- For a subsidy, where $SS^{\text{before subsidy}} \leq SS^{\text{after subsidy}}$, the change in social surplus is $\Delta SS = SS^{\text{after subsidy}} - SS^{\text{before subsidy}}$. Thus, using the identity, we can write

$$\Delta SS + DWL = E,$$

which shows that resources from government expenditures are used to fund the change in social surplus but part of these resources are also being lost in the form of an irrevocable loss. $\square$